



## SUBMITTAL TRANSMITTAL FORM

**MAGUIRE IRON, INC**  
**PO BOX 1446**  
**1610 N MINNESOTA AVE**  
**SIOUX FALLS, SD 57104**  
**PHONE: (605) 334-9749**  
**FAX: (605) 334-9752**

|                  |                                |
|------------------|--------------------------------|
| <b>ENGINEER:</b> | <u>WHKS &amp; Company</u>      |
| <b>OWNER:</b>    | <u>City of Kasson, MN</u>      |
| <b>PROJECT:</b>  | <u>500,000 Gallon Spheriod</u> |
| <b>DATE:</b>     | <u>1/25/2022</u>               |

**SUBMITTAL NO. :** KAS003

| SUBMITTAL DESCRIPTION |
|-----------------------|
| Structural Design     |

### CONTRACTOR'S CERTIFICATION:

Contractor has determined or verified all quantities dimensions field construction criteria, materials catalog numbers and similar data, and has coordinated the information within the submittal with the require-ments of the contract documents, and assumes full responsibility for so doing.

**MAGUIRE IRON, INC**

**By\_ Kaleb Johnson\_\_\_\_\_ Date\_01/25/2022\_\_\_\_\_**

# RVE

*River Valley Engineering, Inc.*

111 North Plaza Court, Van Buren, AR 72956  
(479) 410-2208 Fax (479) 410-2440

Project No. : 2021-743 Subject : Kasson, MN  
Calc. by : JPL Date : 12/20/2021

Page :  
Rev :

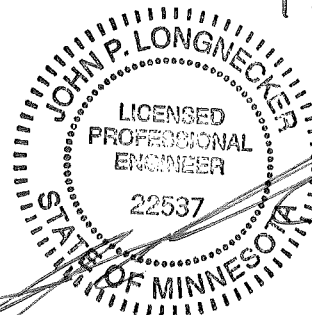
## CALCULATIONS

FOR

500,000 GALLON SPHEROID

WATER STORAGE TANK

KASSON, MINNESOTA



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SPHEROID TANK DESIGN

(Tank Design Conforms to AWWA D-100-2011 Standard)

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|                        |                       |
|------------------------|-----------------------|
| Project Number         | 2021-743              |
| City and State         | Kasson, MN            |
| Date / Time            | 11/19/2021 2:58:52 PM |
| Design Capacity (Gal)  | 500000                |
| Working Capacity (Gal) | 499701                |

DESIGN PARAMETERS

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|                                         |            |
|-----------------------------------------|------------|
| Height to High Water Line (ft)          | 116.6000   |
| Head Range (ft)                         | 40.0000    |
| Free Board (ft)                         | 4.2548     |
| Snow Load (psf)                         | 50         |
| Wind Speed (mph)                        | 100        |
| Seismic Use Group                       | III        |
| Site Class (A,B,C,D,E,F)                | D          |
| Seismic Importance Factor               | 1.5        |
| Earthquake Response Acc. 0.2 Sec, Ss    | 0.048      |
| Earthquake Response Acc. 1.0 Sec, S1    | 0.033      |
| Longer-Period Ground Motion, TL         | 12         |
| Short Period Site Coefficient, Fa       | 1.6        |
| Long Period Site Coefficient, Fv        | 2.4        |
| Damped Short Period Site Coef., SDS     | 0.05120256 |
| Damped Long Period Site Coef., SD1      | 0.05280264 |
| Natural Period of Tower (Seconds)       | 3.2868     |
| Equivalent Fixed Percentage (%)         | 0.5940     |
| Iterations to Convergence               | 3          |
| P-delta Moment due to Seismic (ft-kips) | 234.6888   |
| P-delta Moment due to Wind (ft-kips)    | 126.6393   |
| Deflection Due to Seismic (in)          | 0.6167     |
| Deflection Due to Wind (in)             | 0.3328     |

DESIGN PARAMETERS (cont...)

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Corrosion Allowance

|                 |        |
|-----------------|--------|
| Wetted (in)     | 0.0000 |
| Non-wetted (in) | 0.0000 |

Design Notes:

1. Seismic and Wind Design is per Worst Case of AWWA D-100, IBC and ASCE-7 methods.
2. Specified Additional Shear Due to Antenna (kips) 0.0000
3. Specified Additional Moment Due to Antenna (ft-kips) 0.0000
4. Miscellaneous Load for platforms, piping etc. (kips) 15.0000

## TANK GEOMETRY

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|                                        |          |
|----------------------------------------|----------|
| Diameter (ft)                          | 54.0000  |
| Overall Height of Tank (ft)            | 120.8548 |
| Radius of Roof (ft)                    | 30.3137  |
| Height of Roof (ft)                    | 8.8787   |
| Radius of Knuckle (ft)                 | 19.0000  |
| Offset of Knuckle (ft)                 | 8.0000   |
| Height of Top Knuckle (ft)             | 13.4350  |
| Height of Bottom Knuckle (ft)          | 13.4350  |
| Height of Upper Cone (ft)              | 14.9706  |
| Diameter at Top of Upper Cone (ft)     | 42.8701  |
| Radius of Transition (ft)              | 5.0000   |
| Height of Transition (ft)              | 3.5355   |
| Diameter at Top of Transition (ft)     | 12.9289  |
| Diameter of Pedestal (ft)              | 10.0000  |
| Height of Pedestal (ft)                | 36.3917  |
| Diameter of Base Cone (ft)             | 30.0000  |
| Height of Base Cone (ft)               | 30.0000  |
| Top of Pedestal to Low Water Line (ft) | 10.0000  |
| Diameter of Access Tube (ft)           | 4.0000   |

# Design of Tank Sections

## Roof

|                                             |           |
|---------------------------------------------|-----------|
| Thickness (in)                              | 0.2500    |
| Weight of Roof (kips)                       | 17.2491   |
| Surface Area of Roof (sqft)                 | 1691.0917 |
| Height to Centroid of Plate (ft)            | 3.3295    |
| Height to Centroid of Projected Area (ft)   | 3.3295    |
| Volume of Water in Roof (cuft)              | 5130.9890 |
| Weight of Water in Roof (kips)              | 320.1737  |
| Height to Centroid of Water (ft)            | 2.3119    |
| Shear Due to Earthquake by Water (kips)     | 1.9018    |
| Shear Due to Earthquake by Plate (kips)     | 0.1025    |
| Moment Due to Earthquake by Water (ft-kips) | 4.3969    |
| Moment Due to Earthquake by Plate (ft-kips) | 0.4549    |

## Top Knuckle

|                                             |            |
|---------------------------------------------|------------|
| Thickness (in)                              | 0.2500     |
| Weight of Top Knuckle (kips)                | 24.0105    |
| Surface Area of Top Knuckle (sqft)          | 2353.9712  |
| Height to Centroid of Plate (ft)            | 6.6002     |
| Height to Centroid of Projected Area (ft)   | 6.4802     |
| Volume of Top Knuckle (cuft)                | 27060.9809 |
| Weight of Water in Top Knuckle (kips)       | 1688.6052  |
| Height to Centroid of Water (ft)            | 6.2527     |
| Shear Due to Earthquake by Water (kips)     | 10.0303    |
| Shear Due to Earthquake by Plate (kips)     | 0.1426     |
| Moment Due to Earthquake by Water (ft-kips) | 92.6643    |
| Moment Due to Earthquake by Plate (ft-kips) | 2.7728     |

## Bottom Knuckle

|                                             |            |
|---------------------------------------------|------------|
| Thickness (in)                              | 0.4375     |
| Weight of Bottom Knuckle (kips)             | 42.0184    |
| Surface Area of Bottom Knuckle (sqft)       | 2353.9712  |
| Height to Centroid of Plate (ft)            | 6.9548     |
| Height to Centroid of Projected Area (ft)   | 3.3295     |
| Volume of Bottom Knuckle (cuft)             | 27060.9809 |
| Weight of Water in Bottom Knuckle (kips)    | 1688.6052  |
| Height to Centroid of Water (ft)            | 7.1824     |
| Shear Due to Earthquake by Water (kips)     | 10.0303    |
| Shear Due to Earthquake by Plate (kips)     | 0.2496     |
| Moment Due to Earthquake by Water (ft-kips) | 325.0144   |
| Moment Due to Earthquake by Plate (ft-kips) | 11.1246    |

DESIGN OF TANK SECTIONS (CONT'D)

Upper Cone

| Design -<br>Section | Thickness<br>(in) | Weight<br>(kips) | Height<br>(ft) |
|---------------------|-------------------|------------------|----------------|
| 1                   | 0.6250            | 21.4156          | 4.9902         |
| 2                   | 0.6875            | 17.3505          | 4.9902         |
| 3                   | 0.8750            | 14.1830          | 4.9902         |

Transition

|                                             |          |
|---------------------------------------------|----------|
| Thickness (in)                              | 1.1875   |
| Weight of Transition (kips)                 | 6.5731   |
| Surface Area of Transition (sqft)           | 135.6681 |
| Height to Centroid of Plate (ft)            | 1.9439   |
| Height to Centroid of Projected Area (ft)   | 1.8439   |
| Volume of Transition (cuft)                 | 332.5716 |
| Weight of Water in Transition (kips)        | 20.7525  |
| Height to Centroid of Water (ft)            | 1.9222   |
| Shear Due to Earthquake by Water (kips)     | 0.1233   |
| Shear Due to Earthquake by Plate (kips)     | 0.0226   |
| Moment Due to Earthquake by Water (ft-kips) | 781.7480 |
| Moment Due to Earthquake by Plate (ft-kips) | 12.9278  |

Access Cone

|                              |        |
|------------------------------|--------|
| Thickness (in)               | .5000  |
| Weight of Access Cone (kips) | 2.2787 |

Access Tube

|                                    |        |
|------------------------------------|--------|
| Thickness of Bottom 15 Feet (in)   | .3750  |
| Thickness of Remaining Height (in) | .3750  |
| Weight of Access Tube (kips)       | 9.4059 |

DESIGN OF TANK SECTIONS (CONT'D)

Pedestal  
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| Design -<br>Section | Thickness<br>(in) | Weight<br>(kips) | Height<br>(ft) |
|---------------------|-------------------|------------------|----------------|
| 1                   | 0.7500            | 7.6906           | 8.0000         |
| 2                   | 0.7500            | 15.3812          | 8.0000         |
| 3                   | 0.7500            | 23.0719          | 8.0000         |
| 4                   | 0.7500            | 30.7625          | 8.0000         |
| 5                   | 0.7500            | 34.9843          | 4.3917         |
| Base Cone           |                   |                  |                |
| -----               |                   |                  |                |

| Design -<br>Section | Thickness<br>(in) | Weight<br>(kips) | Height<br>(ft) |
|---------------------|-------------------|------------------|----------------|
| 1                   | 0.8125            | 2.3419           | 2.0000         |
| 2                   | 0.7500            | 9.6942           | 7.0000         |
| 3                   | 0.6875            | 11.9207          | 7.0000         |
| 4                   | 0.6250            | 13.5955          | 7.0000         |
| 5                   | 0.5625            | 14.7186          | 7.0000         |



## Base Plate

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Base Plate Thickness (in) 1.5000

Note: Use a standard 1 in. of non-shrink grout under the base plate.

## Anchor Bolt -- Mild Steel

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| Anchor Bolt<br>Size<br>(in) | Quantity<br>Req'd | Tension<br>Load<br>(kip) | Minimum<br>Embedment<br>(in) |
|-----------------------------|-------------------|--------------------------|------------------------------|
| 1.50                        | 16                | 26334                    | 58.91                        |
| 1.75                        | 16                | 35511                    | 68.09                        |
| 2.00                        | 16                | 46683                    | 78.32                        |

## Anchor Bolt With Plate Attached (Breakout Procedure)

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| Anchor Bolt<br>Size<br>(in) | Quantity<br>Req'd | Tension<br>Load(Ult)<br>(kip) | Minimum<br>Embedment<br>(in) |
|-----------------------------|-------------------|-------------------------------|------------------------------|
| 1.50                        | 16                | 48500                         | 25.00                        |
| 1.75                        | 16                | 48500                         | 25.00                        |
| 2.00                        | 16                | 48500                         | 25.00                        |

## Anchor Bolt Chair

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| Anchor Bolt<br>Size<br>(in) | Top Plate<br>Thickness<br>(in) | Top Plate<br>Width<br>(in) | Side Plate<br>Thickness<br>(in) | Anchor Chair<br>Height<br>(in) |
|-----------------------------|--------------------------------|----------------------------|---------------------------------|--------------------------------|
| 1.50                        | 1.1250                         | 9.0000                     | 0.5000                          | 12.0000                        |
| 1.75                        | 1.1980                         | 9.0000                     | 0.5000                          | 12.0000                        |
| 2.00                        | 1.2208                         | 9.0000                     | 0.5000                          | 12.0000                        |

## LOADS ON THE FOUNDATION

### Loads From Plate Steel

| Section of Tank                    | Weight<br>(Kips) |
|------------------------------------|------------------|
| Roof                               | 17.2491          |
| Top Knuckle                        | 24.0105          |
| Bottom Knuckle                     | 42.0184          |
| Upper Cone                         | 52.9490          |
| Transition                         | 6.5731           |
| Pedestal                           | 34.9843          |
| Base Cone                          | 52.2708          |
| Base Plate                         | 4.8106           |
| Access Shaft                       | 9.4059           |
| Access Cone                        | 2.2787           |
| <hr/>                              |                  |
| Total Load From Plate Steel        | 246.5505         |
| Loads From Miscellaneous           | 15.0000          |
| Loads From Snow                    | 72.1719          |
| Loads From Water                   | 4304.8314        |
| <hr/>                              |                  |
| Total Load On Foundation From Tank | 4638.5538        |

### Moments and Shears on the Foundation

|                                               |           |
|-----------------------------------------------|-----------|
| Wind Moment at Top of Pedestal (ft-kips)      | 1246.8939 |
| Wind Shear at Top of Pedestal (kips)          | 42.1809   |
| Seismic Moment at Top of Pedestal (ft-kips)   | 794.6758  |
| Seismic Shear at Top of Pedestal (kips)       | 26.6354   |
| <hr/>                                         |           |
| Wind Moment at Top of Foundation (ft-kips)    | 4700.2840 |
| Wind Shear at Top of Foundation (kips)        | 59.9716   |
| Seismic Moment at Top of Foundation (ft-kips) | 2818.2508 |
| Seismic Shear at Top of Foundation (kips)     | 27.1537   |

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RINGWALL FOUNDATION DESIGN

(Foundation Design Conforms to AWWA D-100-11 Specification)

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|                     |                       |
|---------------------|-----------------------|
| Project Number      | 2021-743              |
| City and State      | Kasson, MN            |
| Date / Time         | 11/19/2021 5:55:35 PM |
| Tank Capacity (Gal) | 500000                |

DESIGN PARAMETERS

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|                                             |           |
|---------------------------------------------|-----------|
| Diameter of Ringwall (ft)                   | 30.0000   |
| Depth to Bottom of Footing (ft)             | 12.0000   |
| Weight of Water (kip)                       | 4305.0000 |
| Weight of Steel (kip)                       | 247.0000  |
| Weight of Snow (kip)                        | 73.0000   |
| Wind Moment at Top of Ringwall (ft-kip)     | 4701.0000 |
| Wind Shear at Top of Ringwall (kip)         | 60.0000   |
| Seismic Moment at Top of Ringwall (ft-kip)  | 2819.0000 |
| Seismic Shear at Top of Ringwall (kip)      | 28.0000   |
| Compressive Strength of Concrete (ksi)      | 4.0000    |
| Yield Strength of Reinforcement Steel (ksi) | 60.0000   |
| Unit Weight of Soil (pcf)                   | 100.0000  |
| Depth of Water Table (ft)                   | 22.0000   |
| Bearing Capacity for Dead Loads (ksf)       | 3.5000    |
| Bearing Capacity for Wind Loads (ksf)       | 4.2000    |
| Bearing Capacity for Seismic Loads (ksf)    | 4.2000    |
| Coefficient of Passive Pressure             | 2.0000    |
| Coefficient of Friction Between End & Soil  | 0.4000    |

# RINGWALL DESIGN

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|                                            |         |
|--------------------------------------------|---------|
| Diameter C-C of Ringwall (ft)              | 30.0000 |
| Width of Ringwall (ft)                     | 1.6667  |
| Height of Ringwall (ft)                    | 10.2500 |
| Projection of Ringwall Above Grade (ft)    | 0.5000  |
| Radius to Outside of Ringwall (ft)         | 15.8333 |
| Radius to Inside of Ringwall (ft)          | 14.1667 |
| Total Area of Vertical Steel (sqin)        | 37.6991 |
| Area of Vertical Steel on Each Face (sqin) | 18.8496 |
| Total Area of Hoop Steel (sqin)            | 4.9201  |
| Area of Hoop Steel on Each Face (sqin)     | 2.4600  |

## FOOTING DESIGN

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|                                               |          |
|-----------------------------------------------|----------|
| Outside Diameter of Footing (ft)              | 43.1875  |
| Inside Diameter of Footing (ft)               | 10.0000  |
| Width of Footing (ft)                         | 16.5938  |
| Depth to Bottom of Footing (ft)               | 12.0000  |
| Thickness of Footing at Ringwall (ft)         | 2.2500   |
| Thickness of Footing at Edge (ft)             | 1.0000   |
| Total Area of Top Radial Steel (sqin)         | 11.2209  |
| Total Area of Bottom Radial Steel (sqin)      | 130.9001 |
| Total Area of Hoop Steel (sqin)               | 9.6775   |
| Area of Hoop Steel on Each Face (sqin)        | 4.8387   |
| Bearing Stress Due to D.L. Water and Snow     | 3.4729   |
| Bearing Stress Due to D.L. Water and Wind     | 4.0576   |
| Bearing Stress Due to D.L. Water and Seismic  | 3.7969   |
| Safety Factor Due to Overturning form Wind    | 9.2428   |
| Safety Factor Due to Overturning from Seismic | 15.6376  |

FOUNDATION QUANTITIES

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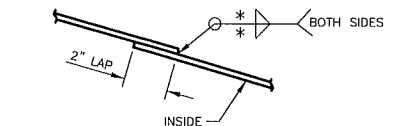
|                                            |            |
|--------------------------------------------|------------|
| Quantity of Concrete in Ringwall (cuyd)    | 59.6333    |
| Quantity of Concrete in Footing (cuyd)     | 91.5657    |
| Total Quantity of Concrete in Found.(cuyd) | 151.1990   |
| Quantity of Ringwall Steel (lb)            | 3867.4320  |
| Quantity of Footing Steel (lb)             | 10776.0613 |
| Total Quantity of Foundation Steel (lb)    | 14643.4933 |

TANK DESIGN DATA

|                                  |                                                                                                                                                                                |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CAPACITY                         | 500,000 GALLONS                                                                                                                                                                |
| SNOW LOAD                        | 50 PSF                                                                                                                                                                         |
| WIND LOAD                        | 100 MPH                                                                                                                                                                        |
| SEISMIC                          | AWWA-D-100-11, IBC                                                                                                                                                             |
| STANDARD                         | AWWA D100-11, IBC                                                                                                                                                              |
| MATERIALS                        | STEEL PLATE-ASTM A36<br>ANCHOR RODS-ASTM A36 OR ASTM F1554 Gr. 36                                                                                                              |
| CORROSION ALLOWANCE              | NONE                                                                                                                                                                           |
| LIGHTING, FIXTURES, AND CONTROLS | PER PROJECT SPECIFICATIONS                                                                                                                                                     |
| RADIOGRAPHIC EXAMINATION         | SPOT X-RAY PER AWWA D100-11                                                                                                                                                    |
| PAINTING AND DISINFECTION        | PER PROJECT SPECIFICATIONS                                                                                                                                                     |
| INSULATION                       | PER PROJECT SPECIFICATIONS                                                                                                                                                     |
| TESTING                          | PER AWWA D100-11                                                                                                                                                               |
| PROJECT SPECIFICATIONS           | SPECIFICATION FOR 500,000 GALLON ELEVATED WATER STORAGE TANK FOR THE CITY OF KASSON, MN BY WHKS WITH GEOTECHNICAL REPORT BY AMERICAN ENGINEERING TESTING DATED: NOVEMBER, 2021 |

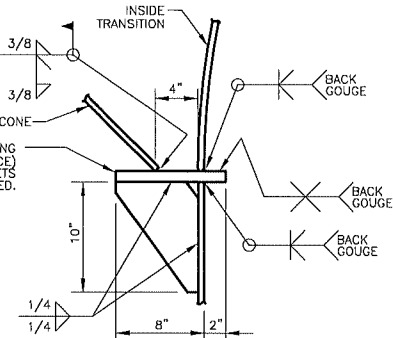
ALL BUTT WELDED JOINTS SUBJECT TO PRIMARY STRESSES THAT ARE DUE TO WATER PRESSURE OR WEIGHT MUST BE FULL PENETRATION AND FULL FUSION.

ALL COMPRESSION AND TENSION RING RADIAL WELDS ARE TO BE FULL PENETRATION AND FULL FUSION.

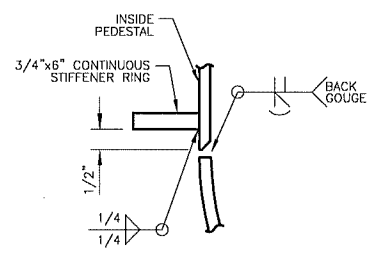


\* FILLET WELD SIZE SHALL BE EQUAL TO THICKNESS OF THINNER OF MEMBERS JOINED.

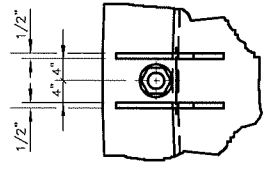
5 E1 DETAIL



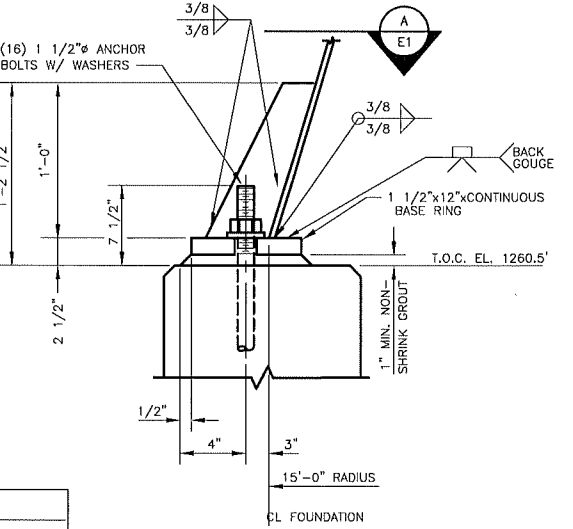
6 E1 DETAIL



7 E1 DETAIL



A E1 DETAIL



8 E1 DETAIL

CONTRACTOR'S CERTIFICATION:

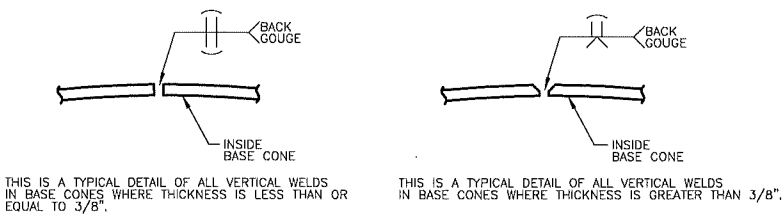
Contractor has determined or verified all quantities, dimensions, field construction criteria, materials, catalog numbers, and similar data, and has coordinated the information within the submittal with the requirements of the contract documents, and assumes full responsibility for so doing.

MAGUIRE IRON, INC.

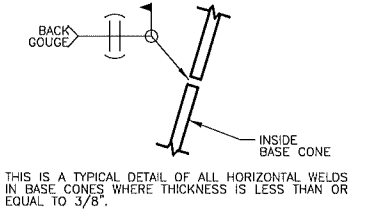
Date \_\_\_\_\_

|                    |                                    |
|--------------------|------------------------------------|
| STRUCTURAL DESIGN  |                                    |
| DATE 1-25-22       |                                    |
| RVE DWG. FILE NAME | 2:\MAGUIRE\2021-743\2021-743E1.DWG |
| SHEET              | 1 OF 2                             |
| RVE PROJECT NO.    | 2021-743                           |
| DRAWING NO.        | 2021-743E1                         |

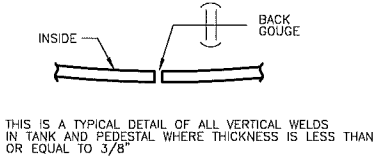
TANK ELEVATION AND DETAILS  
500,000 GALLON SPHEROID  
KASSON, MN



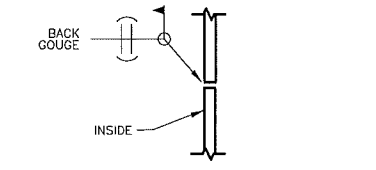
1 E1 DETAIL



2 E1 DETAIL



3 E1 DETAIL



4 E1 DETAIL

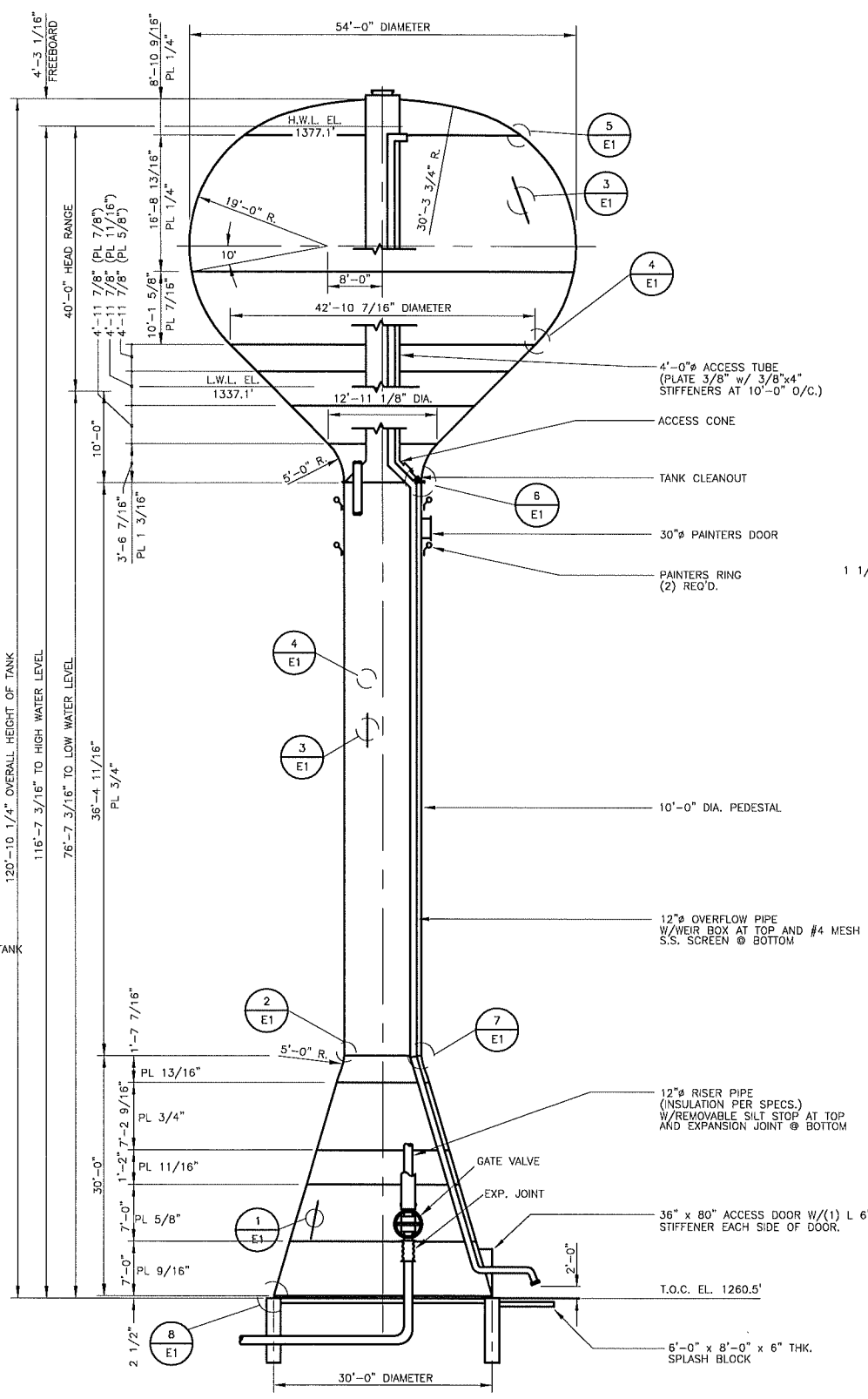
THIS IS A TYPICAL DETAIL OF ALL VERTICAL WELDS IN BASE CONES WHERE THICKNESS IS GREATER THAN 3/8".

THIS IS A TYPICAL DETAIL OF ALL HORIZONTAL WELDS IN TANK AND BASE CONE WHERE THICKNESS IS GREATER THAN 3/8".

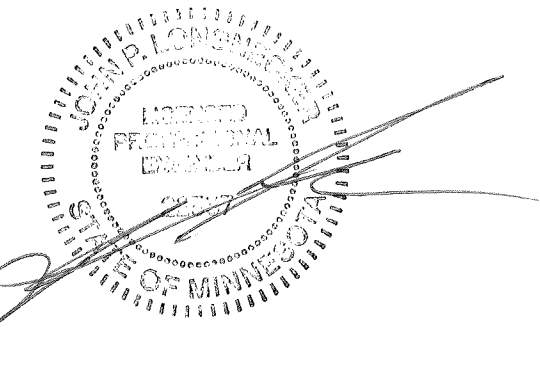
THIS IS A TYPICAL DETAIL OF ALL VERTICAL WELDS IN TANK AND PEDESTAL WHERE THICKNESS IS GREATER THAN 3/8".

THIS IS A TYPICAL DETAIL OF ALL HORIZONTAL WELDS IN TANK AND PEDESTAL WHERE THICKNESS IS GREATER THAN 3/8".

\*\* WHEN THE DIFFERENCE IN THICKNESSES BETWEEN ADJACENT HORIZONTALLY WELDED PLATES EXCEEDS 1/16", A 3:1 TAPERING SHALL BE PERFORMED BETWEEN THE PLATES. THIS TAPERING MAY BE ACHIEVED BY TAPERING THE WELD OR BY TAPERING THE THICKER PLATE.



ELEVATION



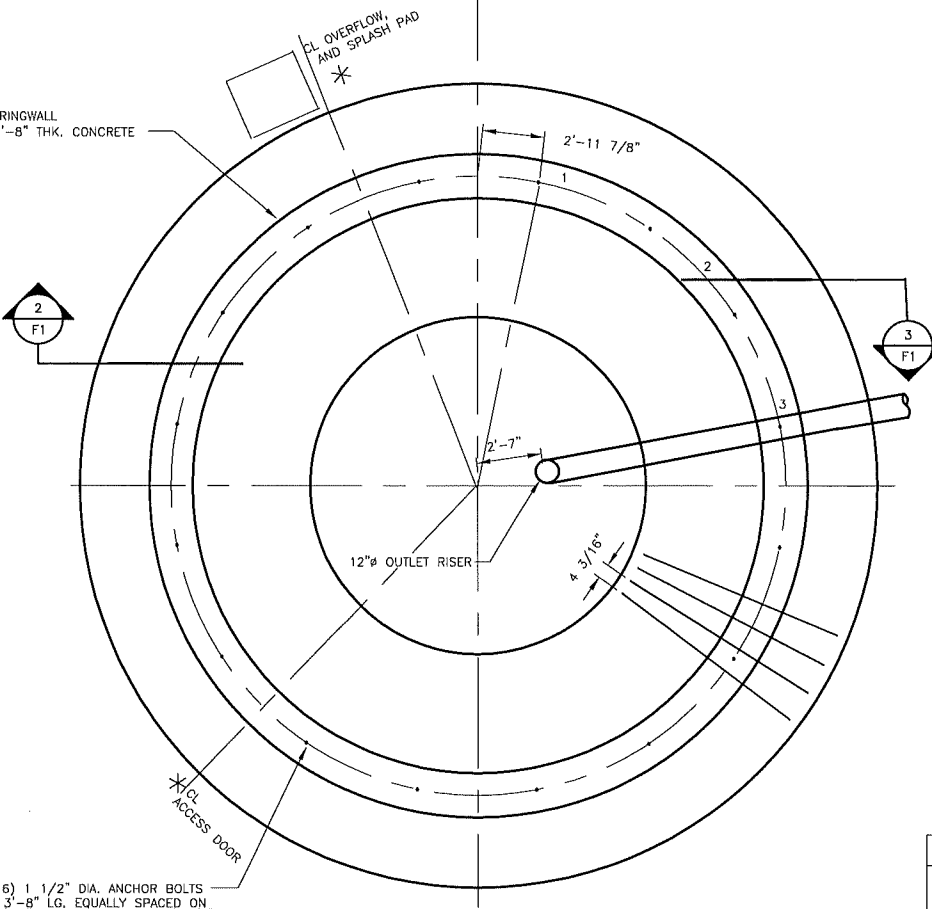
I HEREBY CERTIFY THAT THIS PLAN, SPECIFICATION OR REPORT WAS PREPARED BY ME OR UNDER MY DIRECT SUPERVISION AND THAT I AM A DULY REGISTERED PROFESSIONAL ENGINEER UNDER THE LAWS OF THE STATE OF MINNESOTA.

| NO. | DATE     | DR. BY | CK. BY | DESCRIPTION | SCALE | NONE | DATE     |
|-----|----------|--------|--------|-------------|-------|------|----------|
| 1   | 12-17-21 | ZDL    | JPL    |             |       |      | 12-17-21 |
| 2   | 12-17-21 | JPL    |        |             |       |      | 12-17-21 |

River Valley Engineering, Inc.  
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Fax: (479) 410-2440  
111 North Plaza Ct.  
Van Buren, AR 72956

Maguire Iron, Inc.  
GENERAL CONTRACTOR  
SIOUX FALLS, SOUTH DAKOTA

TANK ELEVATION AND DETAILS  
500,000 GALLON SPHEROID  
KASSON, MN



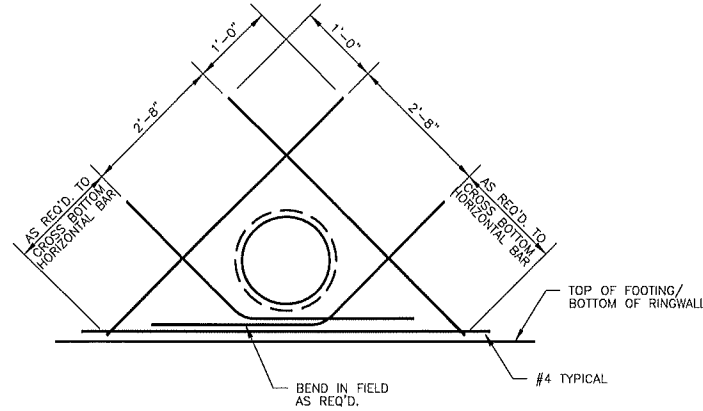
(16) 1 1/2" DIA. ANCHOR BOLTS x 3'-8" LG. EQUALLY SPACED ON 21'-6" DIA. BOLT CIRCLE w/ 3/4" x 6" x 6" PLATE AT BOTTOM END. ANCHOR BOLTS TO BE FURNISHED BY MAGUIRE IRON AND INSTALLED BY

1 FOUNDATION PLAN

| ANCHOR BOLT CHORD DIMENSIONS |             |              |              |
|------------------------------|-------------|--------------|--------------|
| 1 TO 2                       | 1 TO 3      | 1 TO 4       | 1 TO 5       |
| 5'-11 3/8"                   | 11'-8 1/16" | 16'-11 5/16" | 21'-6 13/16" |

PANEL BOX, ELECTRIC BRACKETS AND ELECTRICAL PIPE SLEEVES ARE TO BE LOCATED AND IDENTIFIED BY PROJECT ENGINEER, CONTRACTOR AND APPLICABLE SUBCONTRACTORS.

FOUNDATION CONTRACTOR TO PROVIDE SLEEVES IN FOUNDATION RINGWALL FOR POWER SUPPLY LINE AND TELEPHONE LINE. COORDINATE WITH ELECTRICAL CONTRACTOR AND TELEPHONE COMPANY FOR SIZE AND LOCATION.



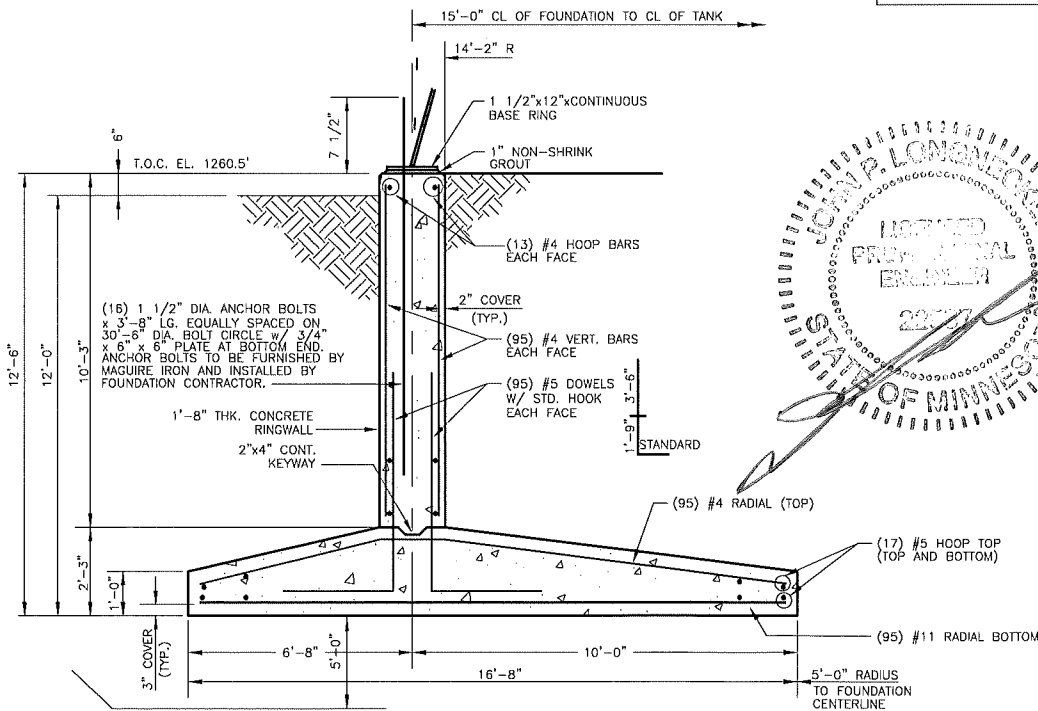
4 DETAIL OF EXTRA REINFORCING AROUND WALL SLEEVE

CONTRACTOR'S CERTIFICATION:

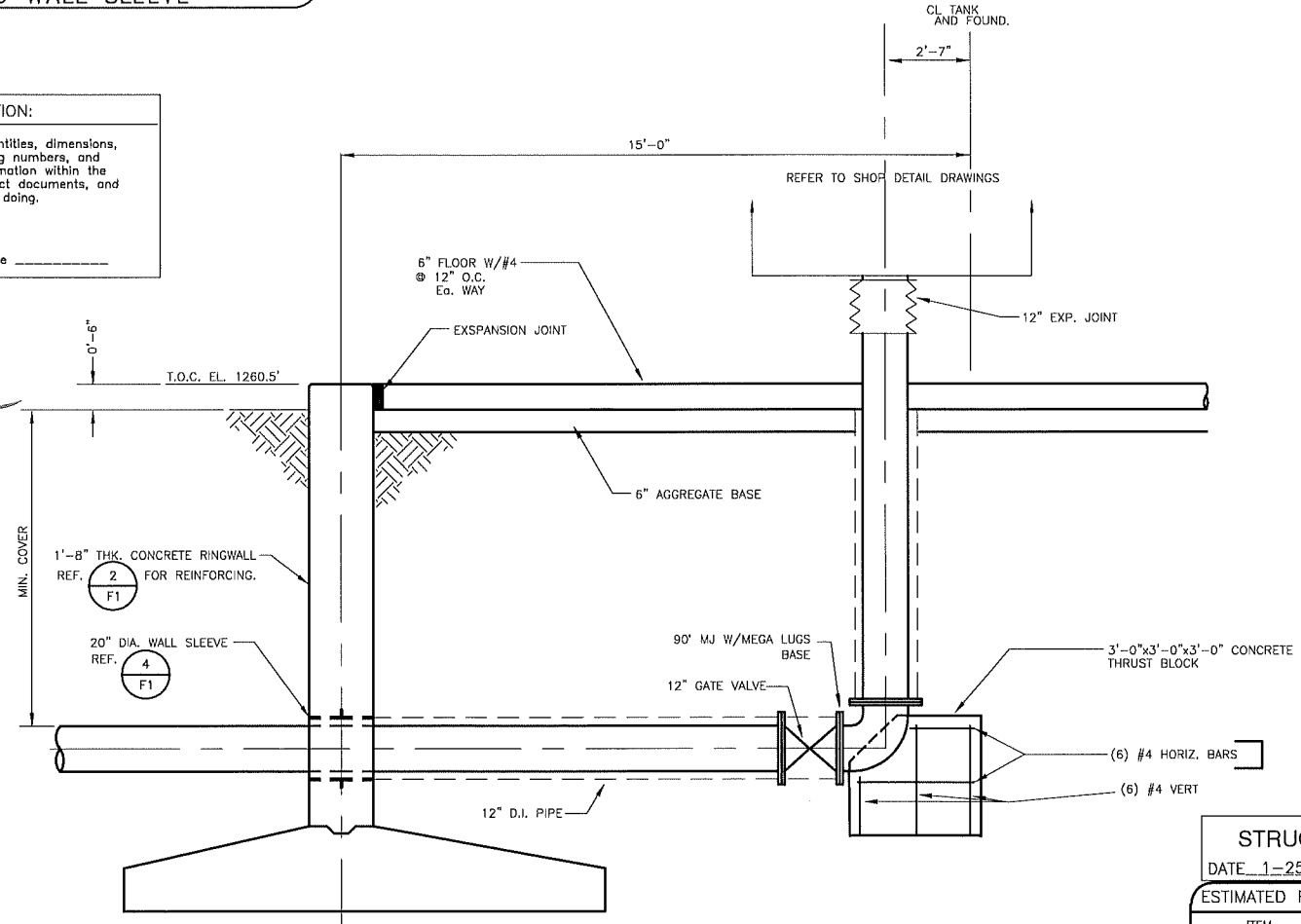
Contractor has determined or verified all quantities, dimensions, field construction criteria, materials, catalog numbers, and similar data, and has coordinated the information within the submittal with the requirements of the contract documents, and assumes full responsibility for so doing.

MAGUIRE IRON, INC.

Date \_\_\_\_\_



2 SECTIONAL ELEVATION THROUGH RINGWALL



3 SECTIONAL ELEVATION AT RISER PIPE

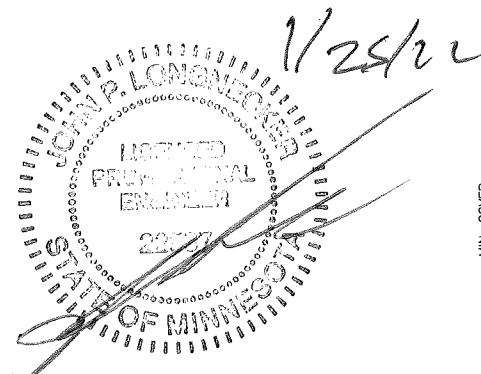
FOUNDATION NOTES

1. ALL FOUNDATION WORK SHALL BE PERFORMED IN ACCORDANCE WITH ACI 318-14 AND

SPECIFICATION FOR 500,000 GALLON ELEVATED WATER STORAGE TANK FOR THE CITY OF KASSON, MN BY WHKS WITH GEOTECHNICAL REPORT BY AMERICAN ENGINEERING TESTING DATED: NOVEMBER, 2021

2. ALL CONCRETE WORK SHALL BE PERFORMED IN ACCORDANCE WITH ACI 318-14.
3. DESIGN AND TOLERANCES SHALL BE PER AWWA D100-11 AND ACI 318-14.
4. ALLOWABLE NET SOIL BEARING PRESSURE FOR WATER, SNOW AND DEAD LOADS=3500 PSF.
5. ALLOWABLE NET SOIL BEARING PRESSURE FOR WATER, DEAD AND WIND LOADS=4200 PSF.
6. ALLOWABLE NET SOIL BEARING PRESSURE FOR WATER, DEAD AND SEISMIC LOADS=4200 PSF.
7. REINFORCING STEEL: ASTM A615 GR. 60. (Fy=60,000)
8. REBAR SPLICE SHALL COMPLY WITH ACI 318-14 w/MIN. OF 42 BAR DIA.
9. CONCRETE SHALL HAVE 28 DAY COMPRESSIVE STRENGTH OF 4000 PSI.
10. ALL EXPOSED CONCRETE EDGES SHALL HAVE A 1"x45° CHAMFER.
11. ALL EXCAVATION, BACKFILL AND LEVELING SHALL BE PERFORMED PER ABOVE REFERENCED SPEC.

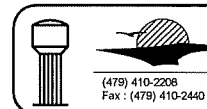
WHKS TO VERIFY ITEMS MARKED WITH (\*).



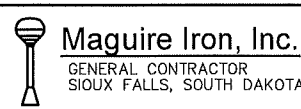
I HEREBY CERTIFY THAT THIS PLAN, SPECIFICATION OR REPORT WAS PREPARED BY ME OR UNDER MY DIRECT SUPERVISION AND THAT I AM A DULY REGISTERED PROFESSIONAL ENGINEER UNDER THE LAWS OF THE STATE OF MINNESOTA.

| NO. | DATE | DR. BY | CK. BY | DESCRIPTION |
|-----|------|--------|--------|-------------|
|     |      |        |        |             |
|     |      |        |        |             |
|     |      |        |        |             |

|       |          |
|-------|----------|
| SCALE | NONE     |
| DR.   | ZDL      |
| DATE  | 12-17-21 |
| CK.   | JPL      |
| DATE  | 12-17-21 |



River Valley Engineering, Inc.  
111 North Plaza Ct.  
Van Buren, AR 72956



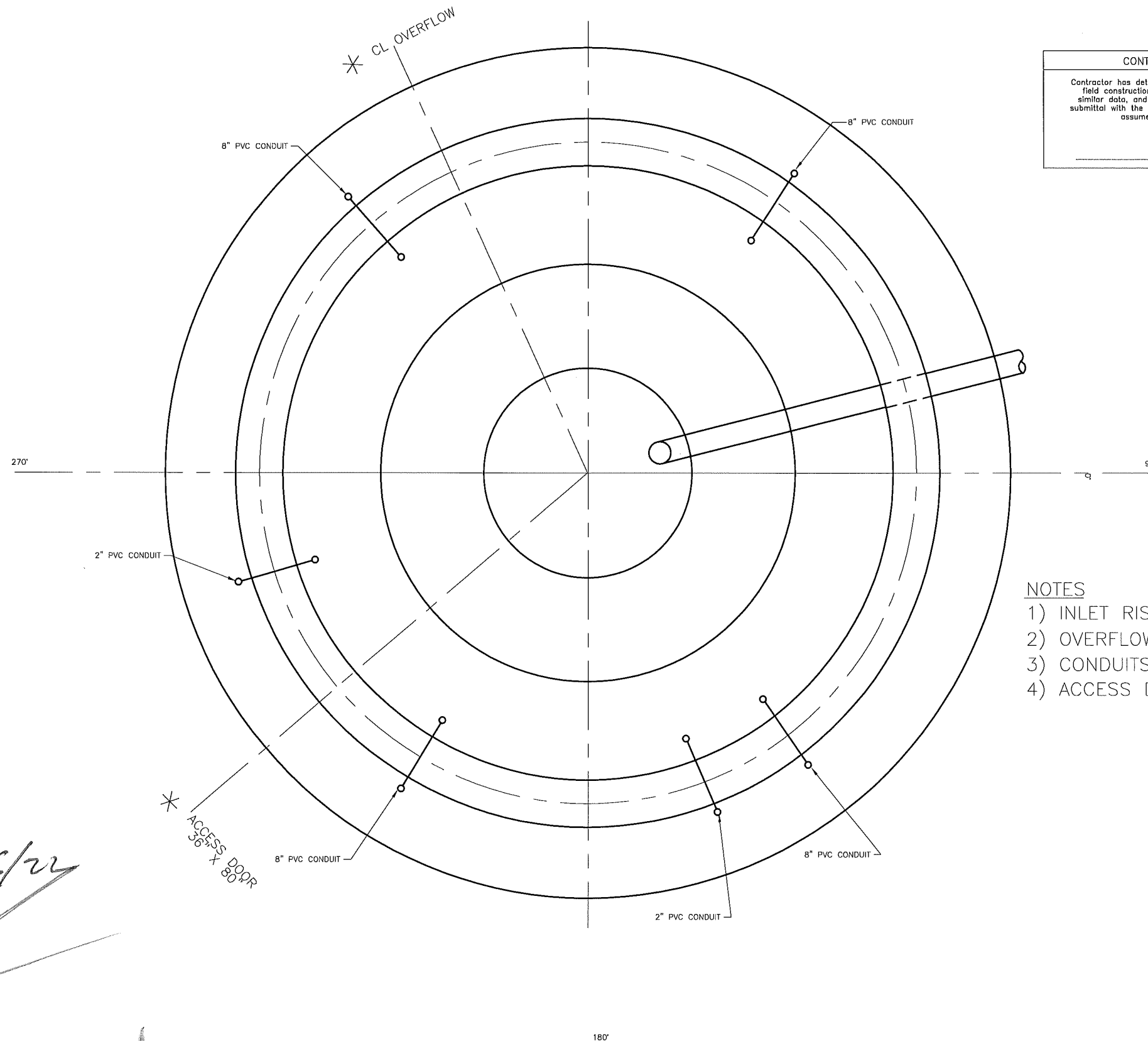
Maguire Iron, Inc.  
GENERAL CONTRACTOR  
SIOUX FALLS, SOUTH DAKOTA

FOUNDATION PLAN AND DETAILS  
500,000 GALLON SPHEROID  
KASSON, MN

| STRUCTURAL DESIGN               |                   |             |
|---------------------------------|-------------------|-------------|
| DATE 1-25-22                    |                   |             |
| ESTIMATED FOUNDATION QUANTITIES |                   |             |
| ITEM                            | CONCRETE (CU YDS) | STEEL (LBS) |
| RINGWALL                        | 60                | 4300        |
| FOOTING                         | 92                | 11900       |
| FLOOR                           | 14                | 1000        |
| PILE CAP                        |                   |             |
| PILE                            |                   |             |
| TOTAL                           | 165               | 17,200      |

|                    |                                    |
|--------------------|------------------------------------|
| RVE DWG. FILE NAME | 2:\MAGUIRE\2021-743\2021-743F1.DWG |
| SHEET              | 2 OF 3                             |
| RVE PROJECT NO.    | 2021-743                           |
| DRAWING NO.        | 2021-743F1                         |





CONTRACTOR'S CERTIFICATION:

Contractor has determined or verified all quantities, dimensions, field construction criteria, materials, catalog numbers, and similar data, and has coordinated the information within the submittal with the requirements of the contract documents, and assumes full responsibility for so doing.

MAGUIRE IRON, INC.

\_\_\_\_\_ Date \_\_\_\_\_

- NOTES
- 1) INLET RISER AT (75°)
  - 2) OVERFLOW AND CATCH BASIN AT (337.5°)
  - 3) CONDUITS (225°)
  - 4) ACCESS DOOR AT (225°)

1/25/22

JOHN P. LONGNECKER  
LICENSED  
PROFESSIONAL  
ENGINEER  
22537  
STATE OF MINNESOTA

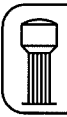
180°

BASE ORIENTATION

I HEREBY CERTIFY THAT THIS PLAN, SPECIFICATION OR REPORT WAS PREPARED BY ME OR UNDER MY DIRECT SUPERVISION AND THAT I AM A DULY REGISTERED PROFESSIONAL ENGINEER UNDER THE LAWS OF THE STATE OF MINNESOTA.

DATE: 1/25/22 REG. NO.: 22537

| NO. | DATE | DR. BY | CK. BY | DESCRIPTION | SCALE | NONE | DATE     |
|-----|------|--------|--------|-------------|-------|------|----------|
|     |      |        |        |             | DR.   | ZDL  | 12-17-21 |
|     |      |        |        |             | CK.   | JPL  | 12-17-21 |



River Valley  
Engineering, Inc.  
(479) 410-2208  
Fax: (479) 410-2440  
111 North Plaza Ct.  
Van Buren, AR 72956



Maguire Iron, Inc.  
GENERAL CONTRACTOR  
SIOUX FALLS, SOUTH DAKOTA



STRUCTURAL DESIGN

DATE: 1-25-22

TANK ORIENTATION  
500,000 GALLON SPHEROID  
KASSON, MN

RVE DWG. FILE NAME  
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|             |            |    |   |                 |          |
|-------------|------------|----|---|-----------------|----------|
| SHEET       | 1          | OF | 2 | RVE PROJECT NO. | 2021-743 |
| DRAWING NO. | 2021-743E1 |    |   |                 |          |